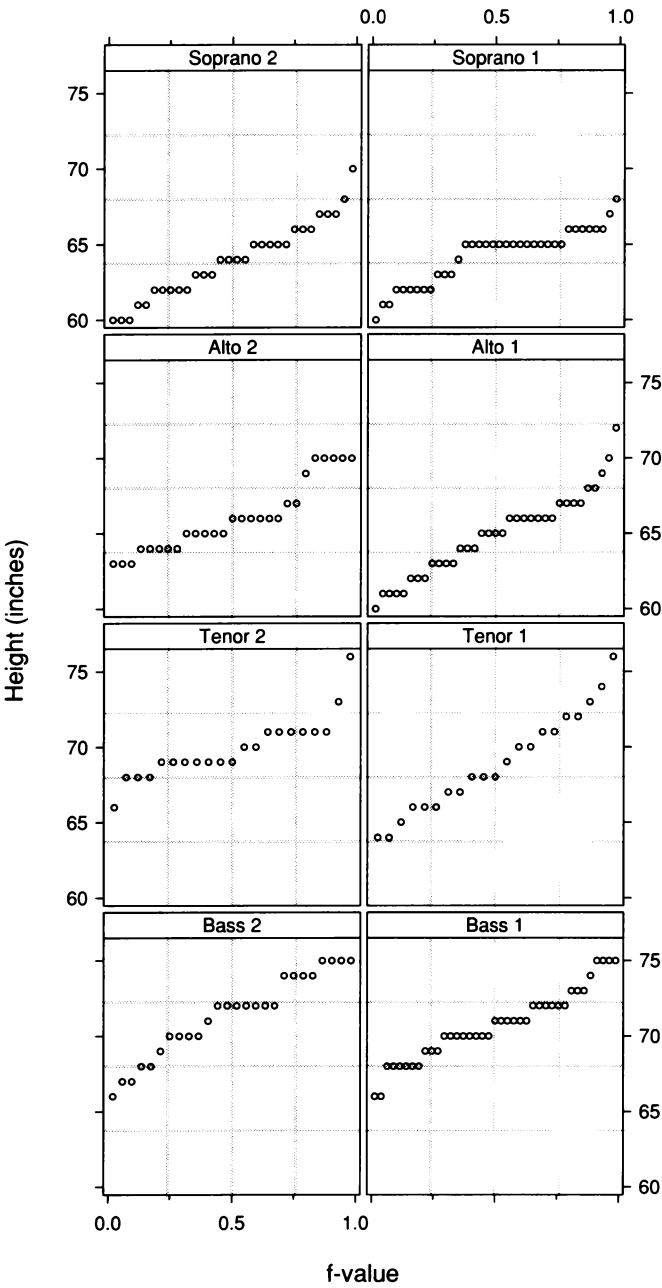




2.1 Quantile plots display univariate data: the heights of singers in the New York Choral Society.

2 *Univariate Data*

Figure 2.1 graphs data introduced in Chapter 1 — heights of singers in the New York Choral Society [16]. The vertical scale on the panels is height; the horizontal scale, which will be explained shortly, ranges from 0 to 1. The singers are divided into eight voice parts, shown by the panel labels in the figure. The altos and sopranos are women and the tenors and basses are men. The pitch intervals of the voice parts are different. Starting from the lower left panel in Figure 2.1, the pitch interval increases as we progress through the panels from left to right and from bottom to top. For example, the second basses must sing lower pitches than the first basses, who in turn sing lower pitches than the second tenors.

The singer heights are *univariate data*: measurements of a single quantitative variable. The measurements are broken up into groups by the categorical variable, voice part. The goal in analyzing the singer data is to determine whether voice part is related to height. One might expect this because taller people tend to be larger overall, and larger vocal tracts would have lower resonance frequencies, and thus produce lower tones.

2.1 *Quantile Plots*

The singer heights for each voice part occupy positions along the measurement scale. The collection of positions is the *distribution* of the data. Thus the goal in analyzing the data is to compare the eight height distributions.

Quantiles

Quantiles are essential to visualizing distributions. The f quantile, $q(f)$, of a set of data is a value along the measurement scale of the data with the property that approximately a fraction f of the data are less

than or equal to $q(f)$. The property has to be approximate because there might not be a value with exactly a fraction f of the data less than or equal to it. The 0.25 quantile is the *lower quartile*, the 0.5 quantile is the *median*, and the 0.75 quantile is the *upper quartile*.

The graphical methods in this chapter are largely visualizations of quantile information. Quantiles provide a powerful mechanism for comparing distributions because f -values provide a standard for comparison. To compare distributions we can compare quantiles with the same f -values. For example, for the singer heights, the median of the second basses is 72 inches, 4 inches greater than the median of the first tenors. This is a meaningful and informative comparison.

An explicit rule is needed for computing $q(f)$. Consider the first tenor heights. Let $x_{(i)}$, for $i = 1$ to n , be the data ordered from smallest to largest; thus $x_{(1)}$ is the smallest observation and $x_{(n)}$ is the largest. For the first tenors, $n = 21$, $x_{(1)} = 64$ inches, and $x_{(21)} = 76$ inches. Let

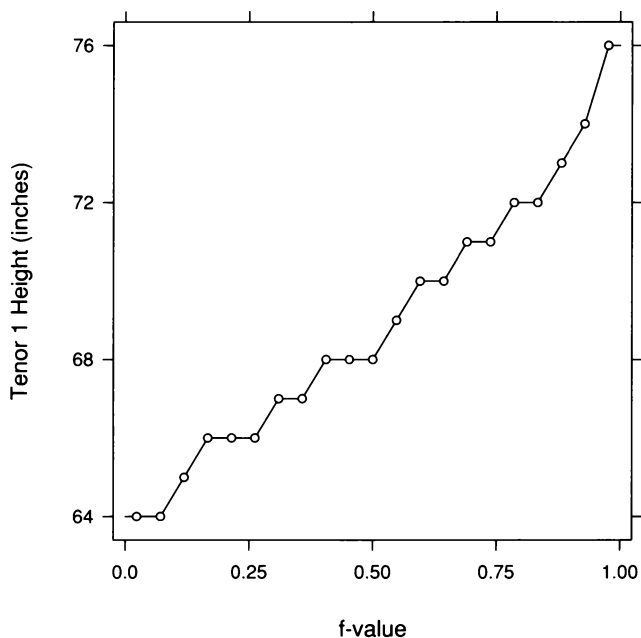
$$f_i = \frac{i - 0.5}{n}.$$

These numbers increase in equal steps of $1/n$ beginning with $1/2n$, which is slightly above zero, and ending with $1 - 1/2n$, which is slightly below one. For the first tenors, the values go from $1/42$ to $41/42$ in steps of $1/21$. We will take $x_{(i)}$ to be $q(f_i)$. For the first tenors the values are

f	x	f	x	f	x
0.02	64	0.36	67	0.69	71
0.07	64	0.40	68	0.74	71
0.12	65	0.45	68	0.79	72
0.17	66	0.50	68	0.83	72
0.21	66	0.55	69	0.88	73
0.26	66	0.60	70	0.93	74
0.31	67	0.64	70	0.98	76

The precise form of f_i is not important; we could have used $i/(n + 1)$, or even i/n , although this last value would prove inconvenient later for visualization methods that employ the quantiles of a normal distribution.

So far, $q(f)$ has been defined just for f -values that are equal to f_i . The definition is extended to all values of f from 0 to 1 by linear interpolation and extrapolation based on the values of f_i and $q(f_i)$. Figure 2.2 illustrates the method using the first tenor heights. The plotting symbols are the points $(f_i, x_{(i)})$; the interpolation and extrapolation are shown by the line segments.



2.2 The symbols and the line segments show the quantile function of the first tenor heights.

Graphing Quantiles

On a *quantile plot*, $x_{(i)}$ is graphed against f_i . In other words, we visualize the f_i quantiles. The panels of Figure 2.1 are quantile plots of the singer heights. The interpolated and extrapolated values of $q(f)$ are not shown because they do not appreciably enhance our visual assessment of the distribution of the data. Rather, the interpolation or extrapolation is used when, for some other purpose, we need a quantile whose f -value does not happen to be one of the values of f_i .

Graphing univariate measurements on a quantile plot is a simple and effective way to have a first look at their distribution. First, the values of all of the data are displayed; we can assess both overall behavior and unusual occurrences. And information about quantiles is conveyed.

Figure 2.1 shows several properties of the singer data: the heights are rounded to the nearest inch; the values for each voice part have a reasonably wide range, about one foot in some cases; there are many first soprano heights equal to 65 inches; and the data have no particularly unusual values, for example, no exceptionally tall or short singers. The height distributions vary substantially with the voice part. At one extreme, the median height is 65 inches for first sopranos, diminutive women piercing the air with notes as high as two octaves above middle C. At the other extreme, the median height is 72 inches for the second basses, tall men vibrating the stage with notes as low as two octaves below middle C. Shortly, other methods for visualizing quantiles will reveal more about the shift in the distributions.

Graphical Order and Visual Reference Grids

Figure 2.1 uses an important convention that will be followed in the remainder of the book; when the panels of a multi-panel display are associated with an ordered variable, such as pitch interval, the variable will increase as we go from left to right and from bottom to top. If the ordered variable were graphed in some way along a horizontal scale, it would increase in going from left to right; if the variable were graphed in some way along a vertical scale, it would increase in going from bottom to top. The *graphical order* of the panels simply follows the established convention.

Figure 2.1 has *visual reference grids*, the vertical and horizontal lines in gray. Their purpose is not to enhance scale reading, or *table look-up*, which is the determination of numerical values from the scales; the tick marks are sufficient for table look-up. Rather, their purpose is to enhance the comparison of patterns, or *gestalts*, on different panels. By providing a common visual reference, the grids enhance our comparison of the relative locations of features on different panels [21]. For example, in Figure 2.1, the grids make it easy to see that almost all second basses are taller than all of the first sopranos.

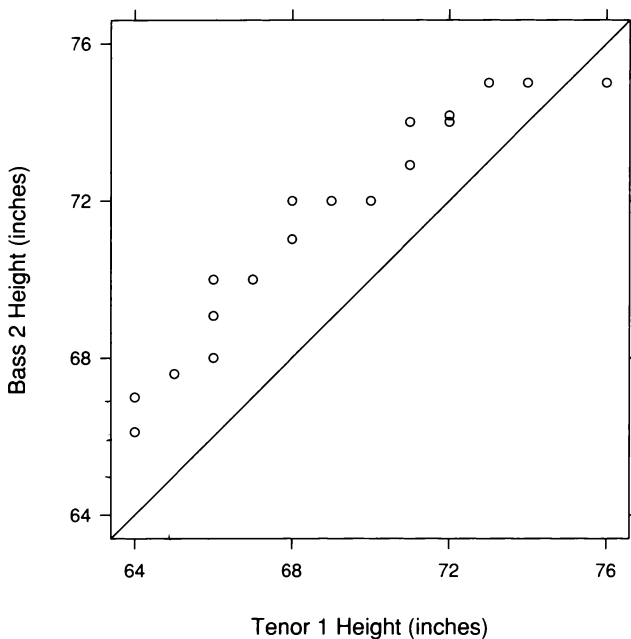
2.2 *Q-Q Plots*

The *quantile-quantile plot*, or *q-q plot*, of Wilk and Gnanadesikan is a powerful visualization method for comparing the distributions of two or more sets of univariate measurements [80]. When distributions are compared, the goal is to understand how the distributions *shift* in going from one data set to the next. For the singers, the goal is to understand how the height distributions shift with voice part.

The most effective way to investigate the shifts of distributions is to compare corresponding quantiles. This was the insightful observation of Wilk and Gnanadesikan, and their invention could not be more simple or elegant — two distributions are compared by graphing quantiles of one distribution against the corresponding quantiles of the other.

Suppose there are just two sets of univariate measurements to be compared. Let $x_{(1)}, \dots, x_{(n)}$ be the first data set, ordered from smallest to largest. Let $y_{(1)}, \dots, y_{(m)}$ be the second, also ordered. Suppose $m \leq n$. If $m = n$, then y_i and x_i are both $(i - 0.5)/n$ quantiles of their respective data sets, so on the q-q plot, $y_{(i)}$ is graphed against $x_{(i)}$; that is, the ordered values for one set of data are graphed against the ordered values of the other set. If $m < n$, then y_i is the $(i - 0.5)/m$ quantile of the y data, and we graph y_i against the $(i - 0.5)/m$ quantile of the x data, which typically must be computed by interpolation. With this method, there are always m points on the graph, the number of values in the smaller of the two data sets. Of course, if m is a big number say 10^3 , then we can select fewer quantiles for comparison.

Figure 2.3 graphs quantiles of the 26 second basses against quantiles of the 21 first tenors. The size of the smaller data set is 21, so 21 quantiles with f-values equal to $(i - 0.5)/21$ are compared. Because some of the plotting symbols overlap, only 18 distinct points appear on the graph. Ordinarily, such overlap would require a remedy discussed in Chapter 3 — jittering, an addition of uniform random noise to coordinates of the points. But on a q-q plot, the points portray what is actually an increasing continuous curve, one quantile function against another, so breaking up the overlap is not necessary. When $i = 1$, the f-value is 0.024. The point in the lower left corner of the data region is the 0.024 quantile for the basses against the 0.024 quantile for the tenors. When $i = 21$, the f-value is 0.976. The point in the upper right corner of the data region is the 0.976 quantile for the basses against the 0.976 quantile for the tenors.



2.3 The first tenor and second bass height distributions are compared by a q-q plot.

The line in Figure 2.3 is $b = t$, where b stands for bass and t stands for tenor. Let (t_i, b_i) be the coordinates of the points graphed on the panel. Our goal in studying the q-q plot is to determine how the points deviate from the line $b = t$. If the distributions of the tenor and bass heights were the same, the points would vary about this line. But they are not the same. There is a shift between the distributions; the underlying pattern of the points is a line

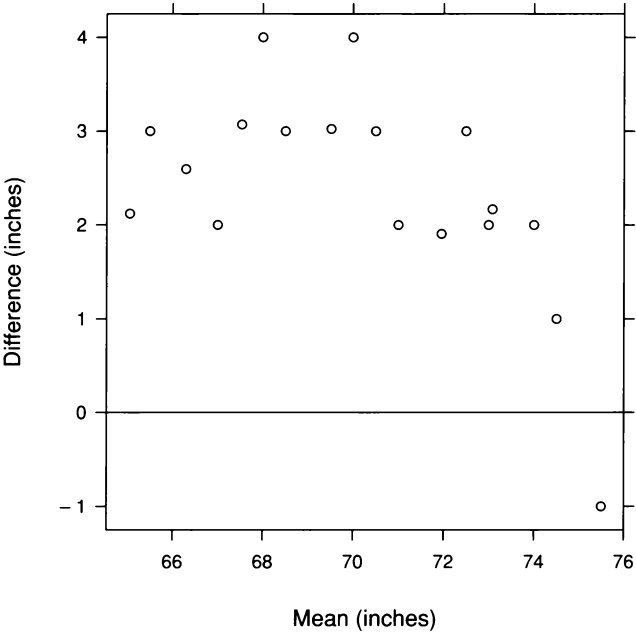
$$b = t + c.$$

Before interpreting this pattern, we will make one more graph to visualize it in a different way.

Tukey Mean-Difference Plots

A *Tukey mean-difference plot*, or *m-d plot*, can add substantially to our visual assessment of a shift between two distributions. Figure 2.4 is an m-d plot derived from the q-q plot in Figure 2.3. The differences, $b_i - t_i$,

are graphed against the means, $(b_i + t_i)/2$. The line $b = t$ on the q-q plot becomes the zero line on the m-d plot, and a shift is assessed by judging deviations from the zero line. This often enhances our perception of effects because we can more readily judge deviations from a horizontal line than from a line with nonzero slope.

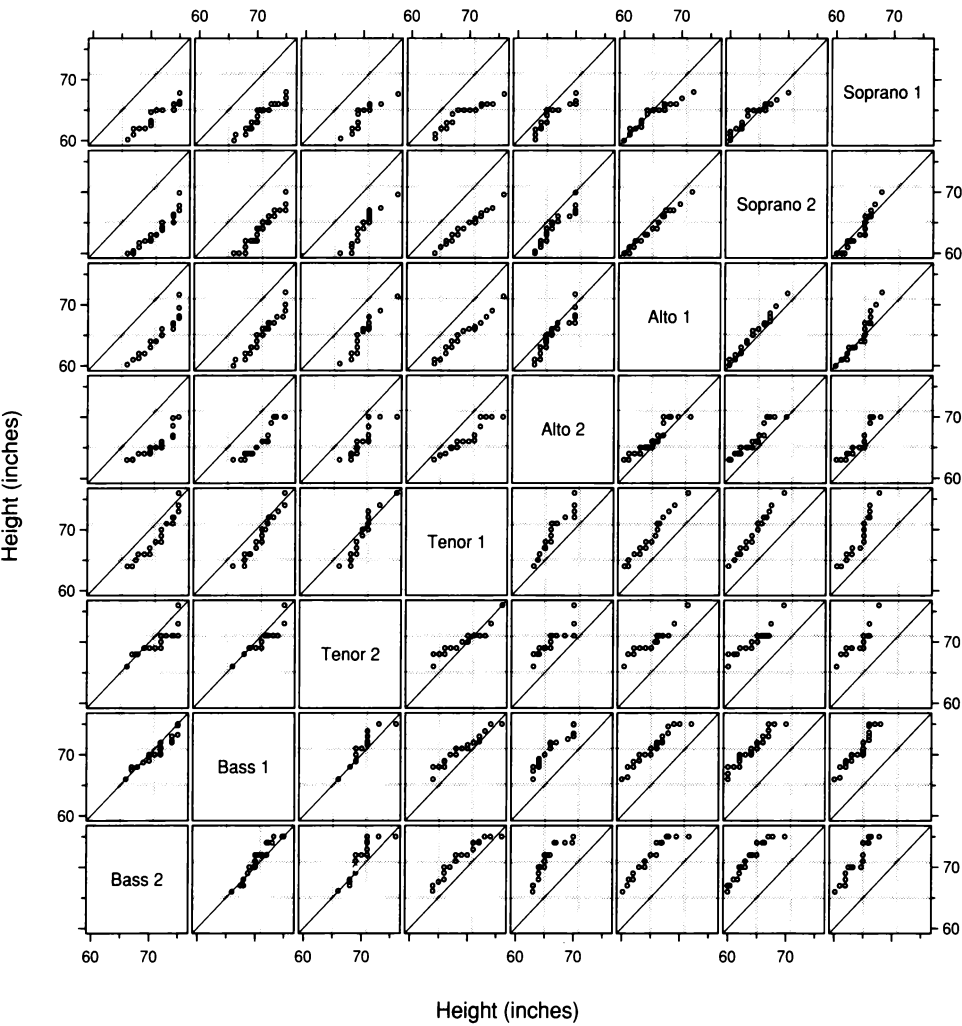


2.4 The first tenor and second bass height distributions are compared by a Tukey m-d plot.

Figures 2.3 and 2.4 show that the tenor and bass distributions differ in an exceedingly simple way: the quantiles of the bass distribution are roughly equal to the quantiles of the tenor distribution plus a constant of about 2.5 inches. There is an *additive shift* of about 2.5 inches. The comparison of the two distributions can be summarized by the simple statement that the distribution of the bass heights is about 2.5 inches greater. This is good news; later examples will show that shifts between distributions can be complex.

Pairwise Q-Q Plots

The goal in analyzing the singer data is to compare the distributions of all voice parts and determine the shifts. Figure 2.5 shows the q-q plots of all possible pairs of voice parts. For example, the second row from the bottom has q-q plots of the first basses against all other voice parts. The second column from the left also compares the distribution of the first basses with all others, but now the first bass quantiles are on the horizontal axis instead of the vertical axis. Thus we can scan either the second row or column to compare the first basses with all other voice parts.

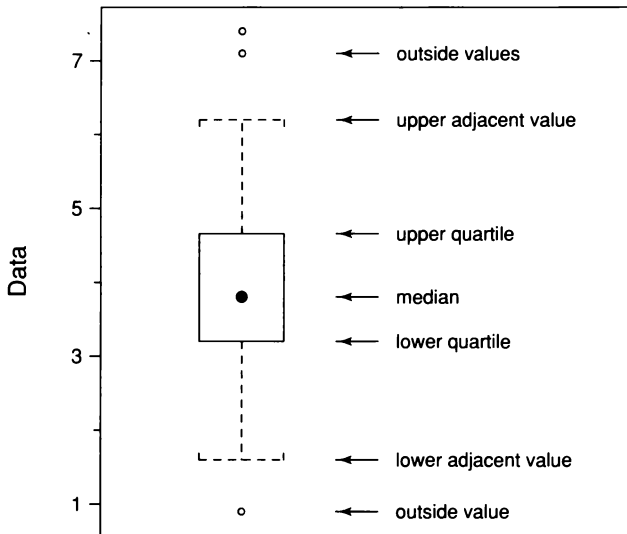


2.5 The height distributions are compared by q-q plots for all pairs of singer voice parts.

There is a great deal of information in Figure 2.5. Overall, there is a suggestion that shifts between pairs of distributions are additive. But we need a way to distill the information because there are so many pairs to compare. Coming methods help with the distillation.

2.3 Box Plots

One method for distilling the information on q-q plots is Tukey's *box plot* [76]. Instead of comparing many quantiles, as on the q-q plot, a limited number of quantities are used to summarize each distribution, and these summaries are compared.

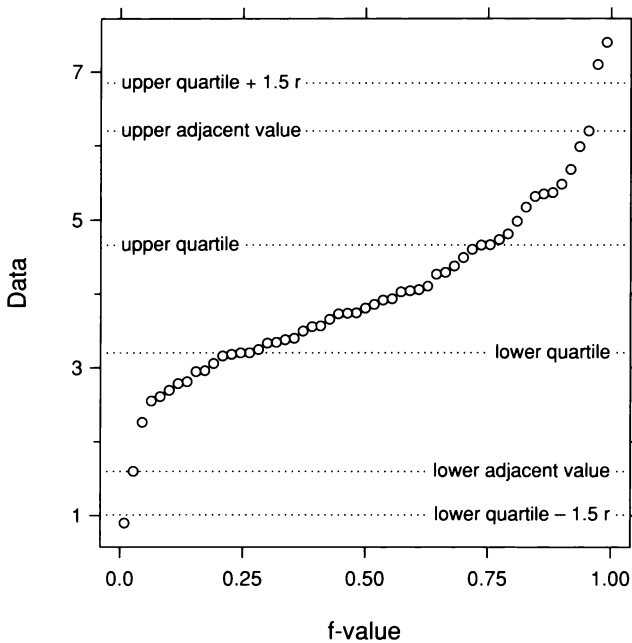


2.6 The diagram defines the box plot display method.

The method of summary is illustrated in Figure 2.6. The filled circle encodes the median, a measure of the center, or *location*, of the distribution. The upper and lower ends of the box are the upper and lower quartiles. The distance between these two values, which is the *interquartile range*, is a measure of the *spread* of the distribution. The middle 50% or so of the data lie between the lower and upper quartiles. If the interquartile range is small, the middle data are tightly packed around the median. If the interquartile range is large, the middle data spread out far from the median. The relative distances of the upper

and lower quartiles from the median give information about the *shape* of the distribution of the data. If one distance is much bigger than the other, the distribution is *skewed*.

The dashed appendages of the box plot encode the *adjacent values*. Let r be the interquartile range. The upper adjacent value is the largest observation that is less than or equal to the upper quartile plus $1.5r$. The lower adjacent value is the smallest observation that is greater than or equal to the lower quartile minus $1.5r$. Figure 2.7, a quantile plot, demonstrates their computation. The adjacent values also provide summaries of spread and shape, but do so further in the extremes, or *tails*, of the distribution.

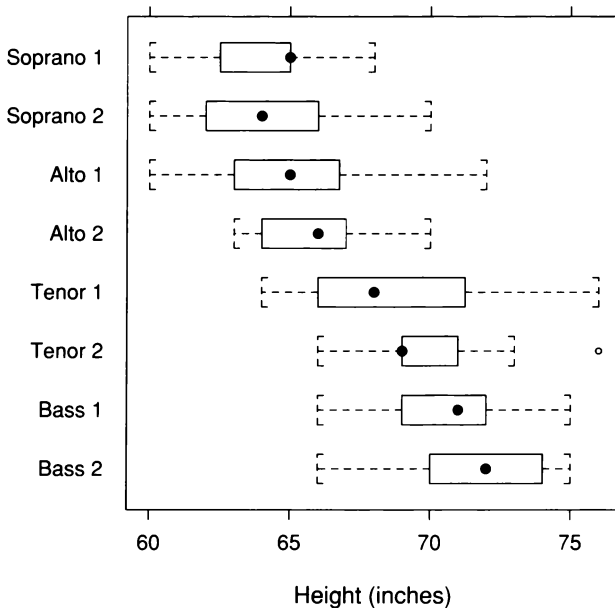


2.7 The diagram illustrates the computation of the adjacent values, which are used in the box plot display method.

Outside values, observations beyond the adjacent values, are graphed individually. Sometimes, the upper adjacent value is the maximum of the data, so there are no outside values in the upper tail; a similar statement holds for the lower tail. Outside values portray behavior in the extreme tails of the distribution, providing further information about

spread and shape. If there happen to be outliers — unusually large or small observations — they appear as outside values, so the box plot method of summarizing a distribution does not sweep outliers under the rug.

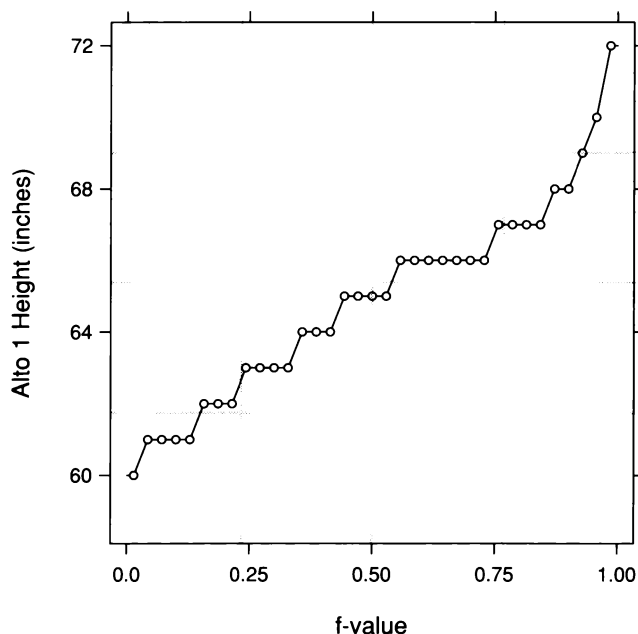
Figure 2.8 shows box plots of the singer heights. The visualization effectively conveys the relationship of voice part and height. Part of the reason is the perceptual effectiveness of the display method. Each specific aspect of the distributions that is encoded — medians, upper quartiles, lower quartiles, and so forth — can be readily visually decoded [21, 77]. For example, it is easy to see the medians as a whole, and visually assess the values. Also, the format is horizontal to enhance the readability of the labels. Figure 2.8 shows that for the men singers, the tenors and basses, height tends to decrease as the pitch interval increases; the same is true of the women singers, the sopranos and altos. One exception to this pattern is the first sopranos, whose height distribution is quite similar to that of the second sopranos. This might be due to a bias in the measurements; the collector of the data noticed a tendency for the shortest sopranos to round their heights strongly upward.



2.8 The eight singer distributions are compared by box plots.

2.4 Normal Q-Q Plots

The box plots in Figure 2.8 show certain effects that accord with our prior knowledge of height and pitch interval, and that we would expect to be reproduced if we were to study another set of singer heights. One such reproducible effect is the general decrease in the height distributions with increasing pitch interval. But the box plots also show spurious effects: unreproducible variation, and variation that is an artifact of rounding to the nearest inch. This spurious variation does not speak to the relationship of height and voice part. For example, for the first sopranos, the median is equal to the upper quartile; for the second tenors, the median is equal to the lower quartile; and the two tallest singers are a first tenor and a second tenor. There is nothing to be learned about height and voice part from these properties of the data. In this section and the next, we move from the graphical methods of the previous sections, which give full vent to the variation in the data, both informative and spurious, to methods whose purpose is to impose structure on the data in an attempt to help us decide what variation appears meaningful and what variation it is better to ignore.

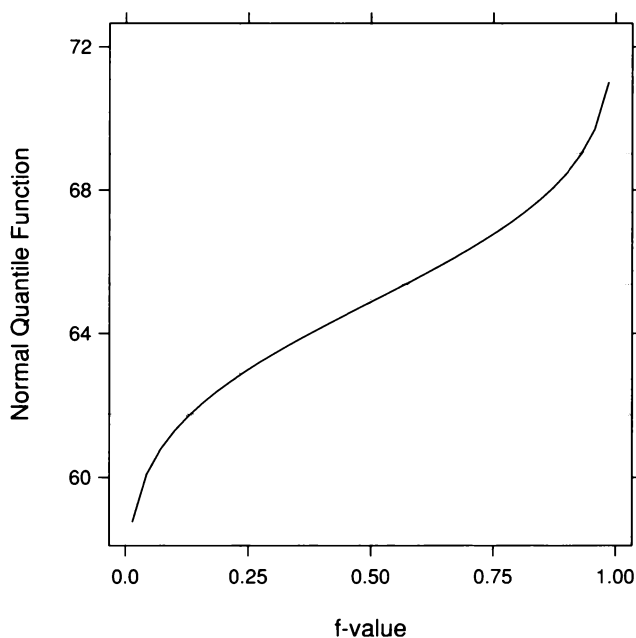


2.9 The plotting symbols and the line segments display the quantile function of the first altos.

Data Quantiles and Normal Quantiles

Figure 2.9 graphs the quantile function, $q(f)$, of the first alto heights. This is the quantile function of a data distribution, a set of real-life univariate measurements. Figure 2.10 graphs quantiles of a mathematical distribution, the normal distribution, a staple of probabilistic inference. It has no reality in the sense that no set of data could ever have such a normal distribution or even be construed as genuinely being a sample from a normal population of potential measurements. It implies, among other things, that measurements range from $-\infty$ to ∞ and have infinite accuracy. Yet it is still helpful to check if the normal quantile function, however imaginary, serves as a reasonable approximation to the real thing, the data distribution.

The normal distribution is a family of distributions. Each normal distribution in the family is specified by two numerical values: the mean, μ , and the standard deviation, σ . The mean is a measure of the location of the distribution, and the standard deviation is a measure of the spread of the distribution about its mean. Given μ and σ , we can



2.10 The curve displays the quantiles of a normal distribution for f -values from $1/70$ to $69/70$, which are the f -values of the minimum and maximum heights of the first altos.

compute the quantile function, $q_{\mu,\sigma}(f)$, of the specified distribution; in Figure 2.10, the specified mean is the sample mean of the first alto heights,

$$\bar{x} = \frac{1}{35} \sum_{i=1}^{35} x_i = 64.9 \text{ inches ,}$$

and the specified standard deviation is the sample standard deviation of these heights,

$$s = \sqrt{\frac{1}{34} \sum_{i=1}^{35} (x_i - \bar{x})^2} = 2.8 \text{ inches .}$$

Most people think about the normal distribution in terms of random variables and probabilities. Suppose nature generates a value, x , of a random variable with a normal distribution. Let f be a probability between 0 and 1. Then the probability that x is less than or equal to $q_{\mu,\sigma}(f)$ is f . That is, a fraction f of the mass of the normal distribution is less than or equal to $q_{\mu,\sigma}(f)$. Notice that the definition of the normal quantile is analogous to the definition of the quantile, $q(f)$, of a set of data; approximately a fraction f of the mass of the data is less than or equal to $q(f)$.

Graphing Data Quantiles and Normal Quantiles

A *normal quantile-quantile plot*, or *normal q-q plot*, is a graphical method for studying how well the distribution of a set of univariate measurements is approximated by the normal. As before, let $x_{(i)}$ be the data, ordered from smallest to largest, and let $f_i = (i - 0.5)/n$. Suppose the distribution of the data is well approximated by some normal distribution with mean μ and standard deviation σ . $x_{(i)}$ is the f_i

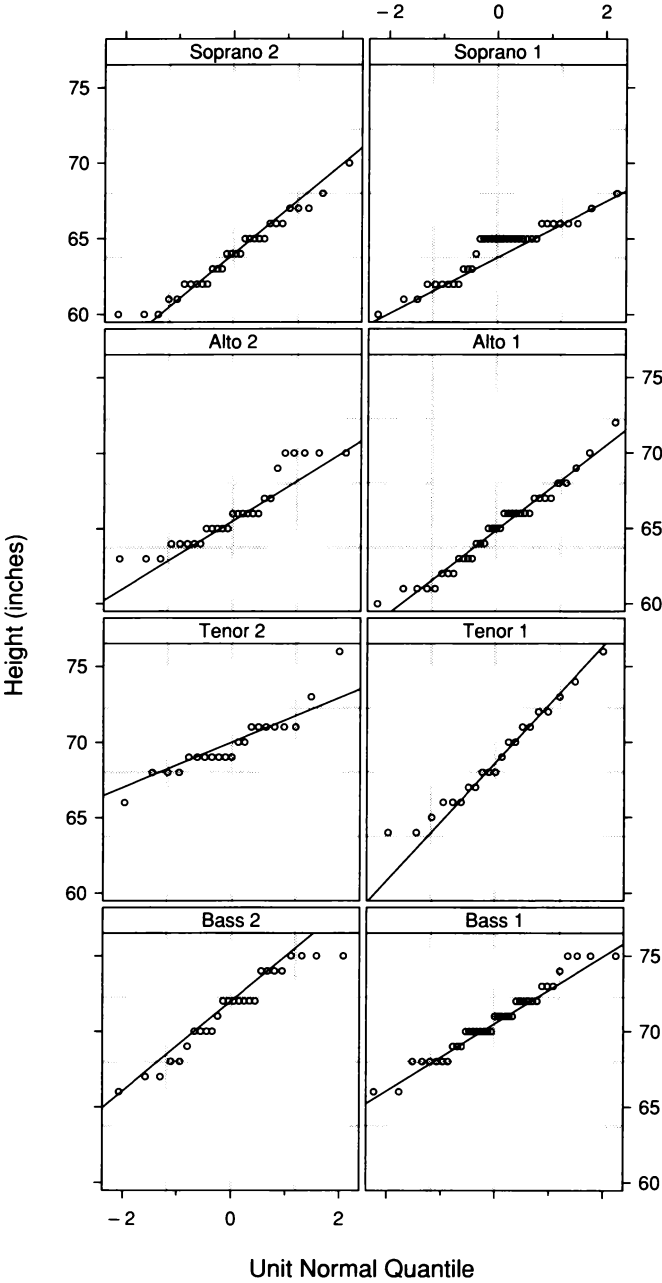
quantile of the data, and $q_{\mu,\sigma}(f_i)$ is the corresponding normal f_i quantile of the approximating distribution. If $x_{(i)}$ is graphed against $q_{\mu,\sigma}(f_i)$, the overall pattern is a line with intercept 0 and slope 1. An important property of normal quantiles is that

$$q_{\mu,\sigma}(f) = \mu + \sigma q_{0,1}(f) ,$$

where $q_{0,1}(f)$ is the quantile function of the unit normal distribution, which has $\mu = 0$ and $\sigma = 1$. In other words, to go from unit normal quantiles to general normal quantiles, we simply multiply by σ and add μ . Thus, if the distribution of the $x_{(i)}$ is well approximated by a normal distribution, then on a graph of $x_{(i)}$ against $q_{0,1}(f_i)$, the overall pattern is a line with intercept μ and slope σ .

A graph of $x_{(i)}$ against $q_{0,1}(f_i)$ is a normal q-q plot. To judge the normal approximation, we judge the underlying pattern of the points on the graph. If the pattern is linear, or nearly so, the data distribution is well approximated by the normal. If not, the deviations from linearity convey important information about how the data distribution deviates from the normal.

Figure 2.11 shows normal q-q plots of the singer data. On each panel, a line is superposed to help us judge the straightness of the pattern. The line passes through the upper and lower quartiles, $(q_{0,1}(0.25), q(0.25))$ and $(q_{0,1}(0.75), q(0.75))$. In each case, the overall pattern appears nearly straight; that is, the eight distributions are reasonably well approximated by normal distributions. The deviations from the overall pattern are inflated by rounding to the nearest inch, which produces strings of points positioned at integer values along the vertical scale. Such discreteness is a departure from normality, because the normal distribution is a continuous one in which any value is possible. But the rounding in this case is not so severe that the approximation is seriously jeopardized.



2.11 Normal q-q plots compare the eight height distributions with the normal distribution.

The Normal Approximation

Many good things happen when data distributions are well approximated by the normal. First, the question of whether the shifts among the distributions are additive becomes the question of whether the distributions have the same standard deviation; if so, the shifts are additive. The slope of the pattern of points on a normal q-q plot is an indicator of the standard deviation of the approximating normal, so judging whether standard deviations are equal from normal q-q plots involves judging whether the slopes are equal. For the singer data, the slopes in Figure 2.11 do vary, but not by a large amount. Most importantly, the variation in the slopes is not related to the means of the distributions. If there were a meaningful change in the standard deviations of the singer distributions, we would expect it to take the form of an increase in the voice-part standard deviations as the voice-part means increase. This is not the case.

A second good happening is that methods of fitting and methods of probabilistic inference, to be taken up shortly, are typically simple and on well understood ground. For example, statistical theory tells us that the sample mean, $\bar{x}$, provides a good estimate of the location of the distribution, and the sample standard deviation, s , provides a good estimate of the spread of the distribution.

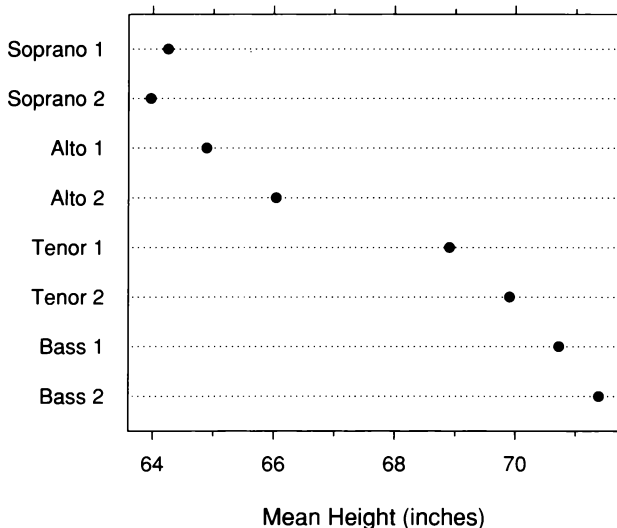
A third good thing is that the description of the data distribution is more parsimonious. A data distribution needs n values to completely determine it, the n observations x_i . If the distribution is well approximated by the normal, these n values can be replaced by two values, $\bar{x}$ and s . The quantiles of the data are then described by the quantiles of the normal distribution with a mean equal to $\bar{x}$ and a standard deviation equal to s . For example, with these values, we know that the upper quartile of the data is about $\bar{x} + 0.67s$.

2.5 Fits and Residuals

Fitting data means finding mathematical descriptions of structure in the data. An additive shift is a structural property of univariate data in which distributions differ only in location and not in spread or shape. For example, the visualization of the singer data has so far suggested that the voice-part distributions differ only in location. An additive shift is fitted by *estimating* location — computing a location measure for each distribution.

Fitting Additive Shifts by Location Estimates

Two candidates for location estimation are the median and the mean. For the singer data, since the normal q-q plots showed that the distributions are well approximated by the normal, we will follow the imperatives of statistical theory and use means. The voice-part means are graphed by a dot plot in Figure 2.12. The mean height decreases with increasing pitch interval, except for the means of the first and second sopranos, which are very nearly equal. We saw this property earlier in the box plots of Figure 2.8, which used medians rather than means as the location measure. At that stage of our analysis, which was preliminary and exploratory, we used medians because they are not distorted by a few outliers, a property not shared by the mean; we will return to this issue of distortion in Section 2.8.



2.12 A dot plot displays the sample means of the height distributions.

Fitted Values and Residuals

For the p th voice part of the singer data, let h_{pi} be the i th measurement of height and let $\bar{h}_p$ be the mean height. In fitting the eight voice-part means to describe the additive shifts, each singer height has had its voice-part mean fitted to it. The *fitted value* for h_{pi} is

$$\hat{h}_{pi} = \bar{h}_p.$$

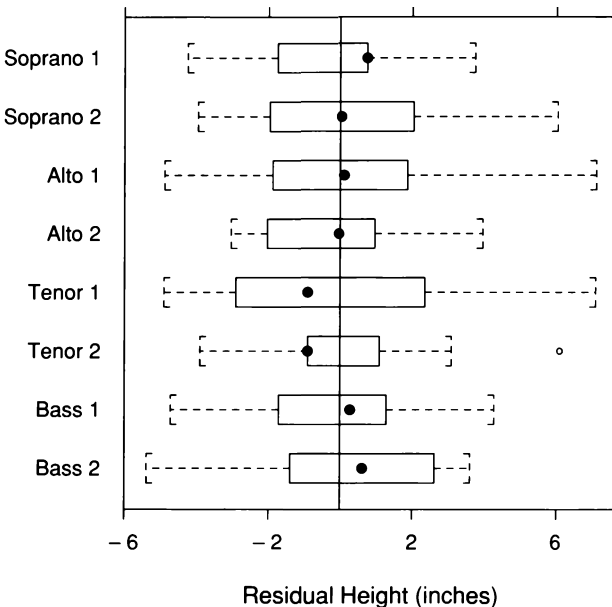
The *residuals* are the deviations of the heights from the fitted values,

$$\hat{\varepsilon}_{pi} = h_{pi} - \hat{h}_{pi}.$$

Thus the heights have been decomposed into two parts,

$$h_{pi} = \hat{h}_{pi} + \hat{\varepsilon}_{pi}.$$

The fitted values account for the variation in the heights attributable to the voice-part variable through the fitting process. The residuals are the remaining variation in the data after the variation due to the shifting means has been removed. This removal is shown in Figure 2.13, which graphs the eight residual distributions. Since the subtraction of means has removed the effect of location, the box plots are centered near zero.



2.13 Box plots compare the distributions of the height residuals for the fit to the data by voice-part means.